

Davide Allegro

📍 Padova, IT ✉ davide.allegro97@gmail.com ☎ +39 340 9158921 📅 Date of birth: 02/06/1997

Work Experience

Postdoctoral Researcher

IAS-Lab, University of Padova, Department of Information Engineering

Padova, IT

Feb 2026 – Present

- **Topics:** 3D Reconstruction and Active Perception
- Supervisor: Prof. Stefano Ghidoni

Teaching Assistant

University of Padova, Department of Information Engineering

Padova, IT

Oct 2022 – Present

- Teaching assistant for the Master's course in Computer Vision.
- Teaching assistant for the Bachelor's course in C++ Programming Laboratory.

AI & Robotics Trainer / Consultant

Danieli Automation

Udine, IT

Nov 2025

- Advanced course on multi-camera hand-eye calibration tailored to company-specific robotic systems.
- Practical course on ROS and Gazebo applied to industrial robotic systems.

Post-Graduate Research Fellow

University of Padova, Department of Information Engineering

Padova, IT

Apr 2022 – Sep 2022

- Research on Multi-Camera Hand-Eye Calibration for human-robot collaboration tasks.
- Contributed to the DrapeBot project on multi-camera calibration and human action recognition.

Education

Ph.D. in Information Engineering

IAS-Lab, University of Padova, Department of Information Engineering

Oct 2022 – Feb 2026

- **Topics:** Multi-Camera to Robot Calibration, 3D Reconstruction and Active Perception for Human-Robot Collaboration
- Supervisor: Prof. Stefano Ghidoni

Visiting PhD Researcher

IMAGINE Lab, Institut Polytechnique de Paris, Paris, France

Apr 2025 – Sep 2025

- **Topics:** Active Mapping of Moving 3D Objects
- Supervisor: Prof. Vincent Lepetit

Master's Degree in Automation Engineering

University of Padova, Department of Information Engineering

Oct 2019 – Apr 2022

- **Thesis:** Automatic Multi-Camera Hand-Eye Calibration for Robotic Workcells.
- Supervisor: Prof. Stefano Ghidoni
- Final Degree Mark: 106/110

Bachelor's Degree in Information Engineering

University of Padova, Department of Information Engineering

Sep 2016 – Sep 2019

- **Thesis:** Neural Networks and Deep Learning
- Supervisor: Prof. Augusto Ferrante
- Final Degree Mark: 101/110

Awards

Manager Anch'io Competition 2024 <i>Awarded for the best thesis in Artificial Intelligence, Automation, and Robotics.</i>	2024
Klaus Fischer Degree Award 2023 <i>Awarded for the best thesis on innovations in processes for the digitalization and automation of industrial production.</i>	2023
ADAPT Field Campaign Competition 2022 <i>First prize in the Advanced Agile Production competition organized by Tampere University, Finland.</i>	2022

Publications

- D. Picard, N. Dufour, L. Degeorge, A. Ghosh, **D. Allegro**, et al., “*PoM: A Linear-Time Replacement for Attention with the Polynomial Mixer*”, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- D. Allegro**, S. Li, S. Ghidoni, V. Lepetit, “*Paparazzo: Active Mapping of Moving 3D Objects*”, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- L. Barcellona, A. Zadaianchuk, **D. Allegro**, S. Papa, S. Ghidoni, E. Gavves, “*Dream to manipulate: Compositional world models empowering robot imitation learning with imagination*”, International Conference on Learning Representations (ICLR), 2025.
- M. Terreran, L. Bragagnolo, **D. Allegro** and S. Ghidoni, “*A Lightweight Ensemble Framework for Online Skeleton-Based Human Action Recognition in Industrial Environments*”, European Conference on Mobile Robots (ECMR), 2025.
- D. Allegro**, M. Terreran and S. Ghidoni, “*MEMROC: Multi-Eye to Mobile Robot Calibration*”, International Conference on Intelligent Robots and Systems (IROS), 2024.
- L. Bragagnolo, M. Terreran, **D. Allegro** and S. Ghidoni, “*Multi-view Pose Fusion for Occlusion-Aware 3D Human Pose Estimation*”, European Conference on Computer Vision (ECCV), 2024.
- A. Bacchin, **D. Allegro**, S. Ghidoni and E. Menegatti, “*Sood-imagenet: a large-scale dataset for semantic out-of-distribution image classification and semantic segmentation*”, European Conference on Computer Vision (ECCV), 2024.
- D. Allegro**, M. Terreran and S. Ghidoni, “*Multi-camera hand-eye calibration for human-robot collaboration in industrial robotic workcells*”, IEEE Robotics and Automation Letters (RA-L), 2024.
- D. Allegro**, M. Terreran and S. Ghidoni, “*Metric—multi-eye to robot indoor calibration dataset*”, Information, 2023.
- D. Evangelista, E. Olivastri, **D. Allegro**, E. Menegatti and A. Pretto, “*A graph-based optimization framework for hand-eye calibration for multi-camera setups*”, International Conference on Robotics and Automation (ICRA), 2023.
- D. Evangelista, **D. Allegro**, M. Terreran, A. Pretto, and S. Ghidoni, “*An unified iterative hand-eye calibration method for eye-on-base and eye-in-hand setups*”, International Conference on Emerging Technologies and Factory Automation (ETFA), 2022.

Languages

English: Professional C1

Italian: Native

Technologies

Languages: Python, C++

Tools: Git, Pytorch, OpenCV, ROS, Gazebo